

Tarangini Workflow Suite

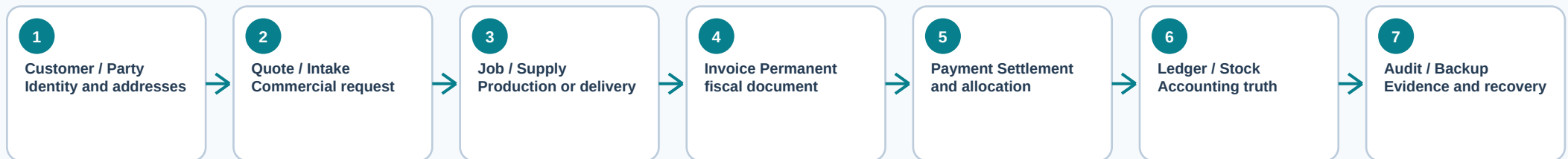
Every Transaction UML and Control Map

Implemented billing, accounting, inventory, customer intake, production, warranty, sync and backup flows - plus clearly marked proposed payment, GST and online rollout designs.

Generated from the current source architecture | 39 transaction diagrams

End-to-End Transaction Spine

OVERVIEW



Rules that connect every transaction

- org_id separates each company and controls access.
- party_id and address snapshots preserve the exact customer/vendor used.
- item_id connects invoices, purchases, stock and job material consumption.
- job_id, bill_id and payment allocations connect production to accounting.
- Financial numbers are never reused after save or soft deletion.
- Audit records identify user, reason, device/time and before/after impact.
- Only one Main System is the accounting and stock source of truth.
- PROPOSED diagrams are designs only and do not claim existing functionality.

Transaction Index - Part 1

[INDEX](#)

01	Login, Session and Role Authorization	Access and organization	IMPLEMENTED	5
02	Company, User and Permission Administration	Masters	IMPLEMENTED	6
03	Party, Shared Organization and Address Book	Masters	IMPLEMENTED	7
04	Item, Service, Barcode and Opening Stock	Masters and inventory	IMPLEMENTED	8
05	Quotation Transaction	Billing	IMPLEMENTED	9
06	Delivery Challan Transaction	Billing	IMPLEMENTED	10
07	Proforma Invoice Transaction	Billing	IMPLEMENTED	11
08	Sale, POS and Project-Printing Invoice	Billing and accounting	IMPLEMENTED	12
09	Invoice Edit, Owner Delete and Invoice Correction	Billing controls	IMPLEMENTED	13
10	Duplicate and Cross-Organization Voucher	Billing productivity	IMPLEMENTED	14
11	Payment Received and Invoice Allocation	Payments	IMPLEMENTED	15
12	Payment Paid / Expense Settlement	Payments	IMPLEMENTED	16
13	Owner Approval for Payment Corrections	Payments	PROPOSED	17
14	Purchase Order to Purchase Voucher	Purchasing	IMPLEMENTED	18
15	Direct Purchase Voucher	Purchasing	IMPLEMENTED	19
16	Expense Voucher	Accounting	IMPLEMENTED	20
17	Credit Note, Sales Return and Refund	Returns and adjustments	IMPLEMENTED	21
18	Debit Note and Purchase Return	Returns and adjustments	IMPLEMENTED	22
19	Manual Journal and Financial-Year Closing	Accounting	IMPLEMENTED	23
20	Bank Statement Import and Reconciliation	Banking	IMPLEMENTED	24

Transaction Index - Part 2

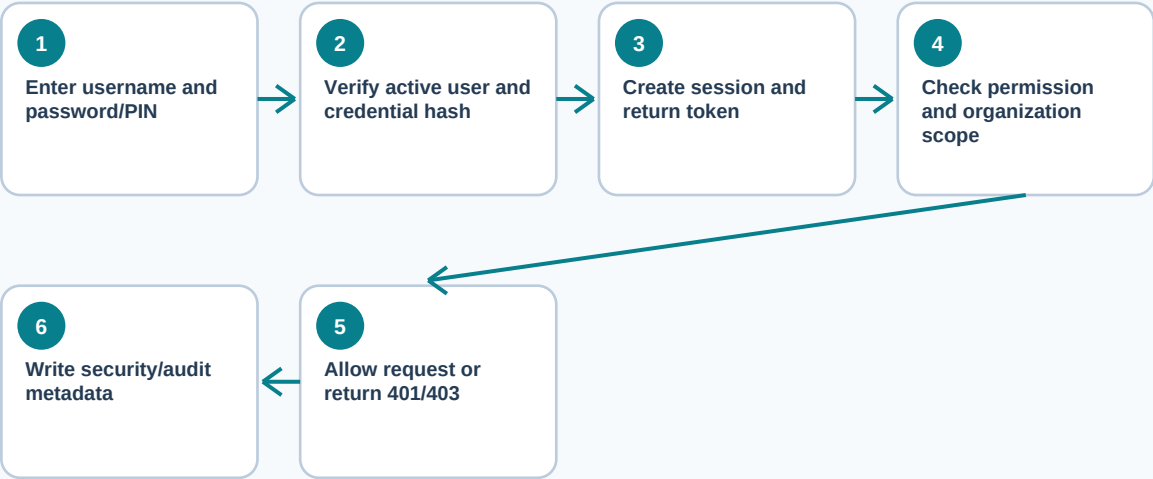
[INDEX](#)

21	GST Reports and GSTR-2B Matching	GST compliance	IMPLEMENTED	25
22	GST E-Invoice Registration	GST compliance	PROPOSED	26
23	GST E-Way Bill Generation and Cancellation	GST compliance	PROPOSED	27
24	Customer QR Intake, Upload and PDF Analysis	Customer intake	IMPLEMENTED	28
25	Intake Review and Job Creation	Job orders	IMPLEMENTED	29
26	Estimate, Quotation and Customer Approval	Job orders	IMPLEMENTED	30
27	Job Assignment and Operator Response	Production	IMPLEMENTED	31
28	Production Status, Work Report and Daily Log	Production	IMPLEMENTED	32
29	Job Material Consumption	Production and inventory	IMPLEMENTED	33
30	Additional Work, Evidence and Customer Decision	Production	IMPLEMENTED	34
31	First Bill, Final Invoice, Delivery and Collection	Job billing and delivery	IMPLEMENTED	35
32	Order Tracking, Quotation and Output Preview	Customer portal	IMPLEMENTED	36
33	Attachment Analysis, Archive and Retention	Files	IMPLEMENTED	37
34	Warranty Replacement and Service-Center Tracking	Warranty	IMPLEMENTED	38
35	Shift, Held Bill and Day-End Acceptance	POS	IMPLEMENTED	39
36	Client Offline Queue and Main-System Synchronization	Network continuity	IMPLEMENTED	40
37	Manual Migration Backup, Encrypted Backup and Restore	Backup and recovery	IMPLEMENTED	41
38	Public Customer Website Transaction	Online rollout	PROPOSED	42
39	Android Customer and Staff Application	Online rollout	PROPOSED	43

Login, Session and Role Authorization

ACTORSOwner, counter, finance, operator, engineer

PURPOSEAuthenticates the user, creates a server session and applies company and role permissions to every protected request.

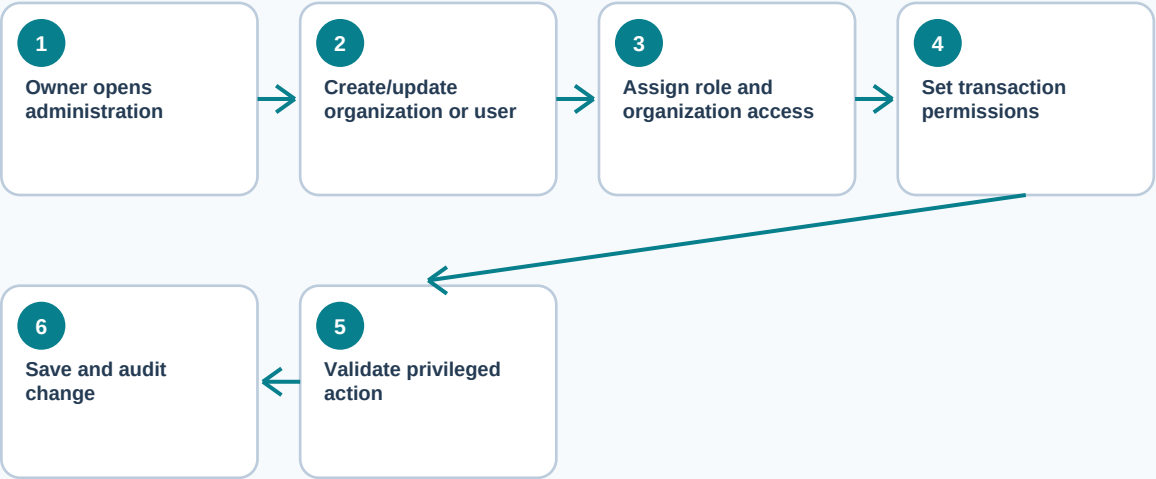


ACCOUNTING EFFECT	STOCK EFFECT	MAIN RECORDS	KEY CONTROLS
None	None	users, sessions, audit_log	Expiry, role permissions, org scope, owner-only actions

Company, User and Permission Administration

ACTORS Owner1

PURPOSE Creates companies and staff accounts, assigns organization access and limits billing, accounting, reporting and job functions.

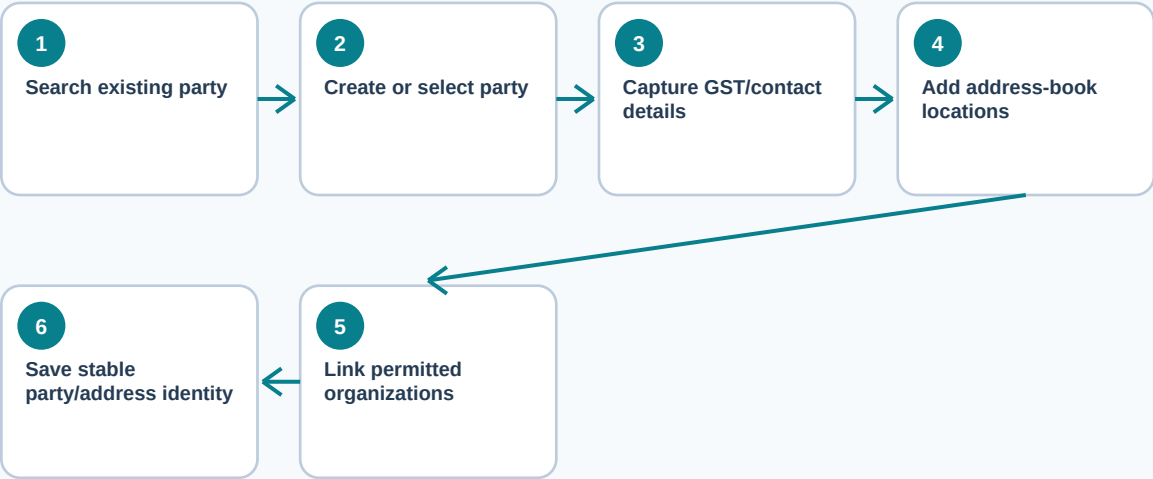


ACCOUNTING EFFECT No direct posting	STOCK EFFECT No direct movement	MAIN RECORDS orgs, users, transaction_control_settings, audit_log	KEY CONTROLS Owner-only administration; no cross-org access without permission
---	---	---	--

Party, Shared Organization and Address Book

ACTORSOwner, permitted billing staff

PURPOSECreates a customer/vendor, optionally shares it with selected companies and stores multiple billing, delivery, branch or site addresses.

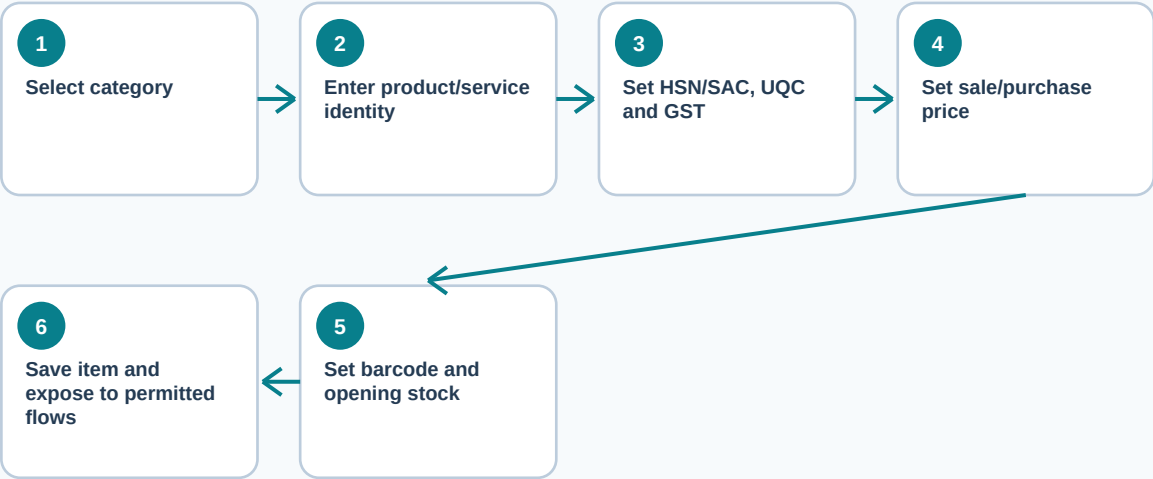


ACCOUNTING EFFECT Opening balance may post receivable/payable	STOCK EFFECT None	MAIN RECORDS parties, party_org_links, party_addresses, journal_entries	KEY CONTROLS Same-name parties distinguished by ID, GSTIN, phone and address
---	-----------------------------	---	--

Item, Service, Barcode and Opening Stock

ACTORSOwner, inventory staff

PURPOSECreates products/services with HSN/SAC, rates, barcode, opening stock and reorder level for billing and job material use.

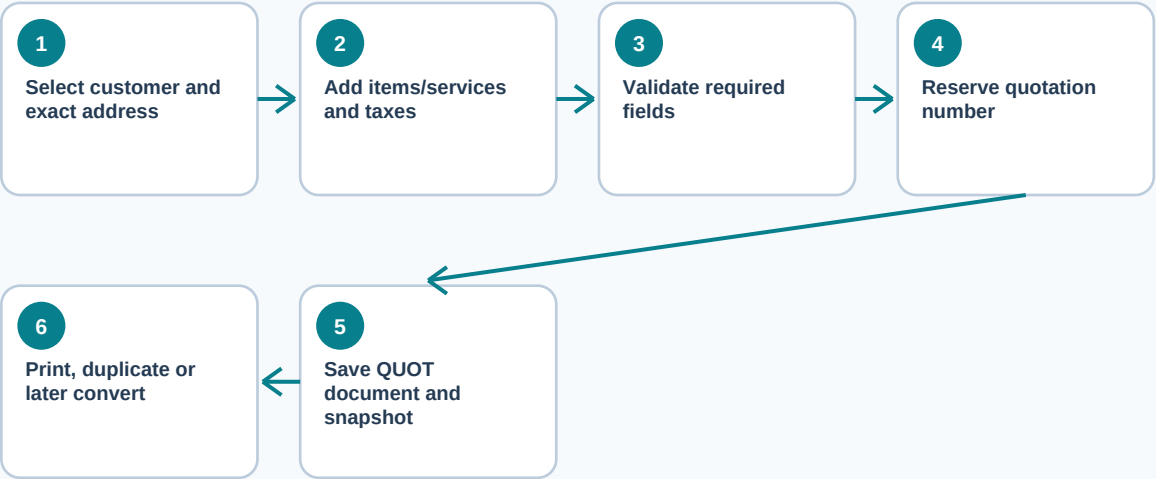


ACCOUNTING EFFECT No direct journal from item master	STOCK EFFECT Opening stock becomes stock-report baseline	MAIN RECORDS item_categories, items	KEY CONTROLS Organization ownership, unique item identity, active/inactive state
--	--	---	--

Quotation Transaction

ACTORS Owner, counter, permitted operator

PURPOSE Records a priced offer using a stable party and address snapshot without posting accounting or reducing stock.

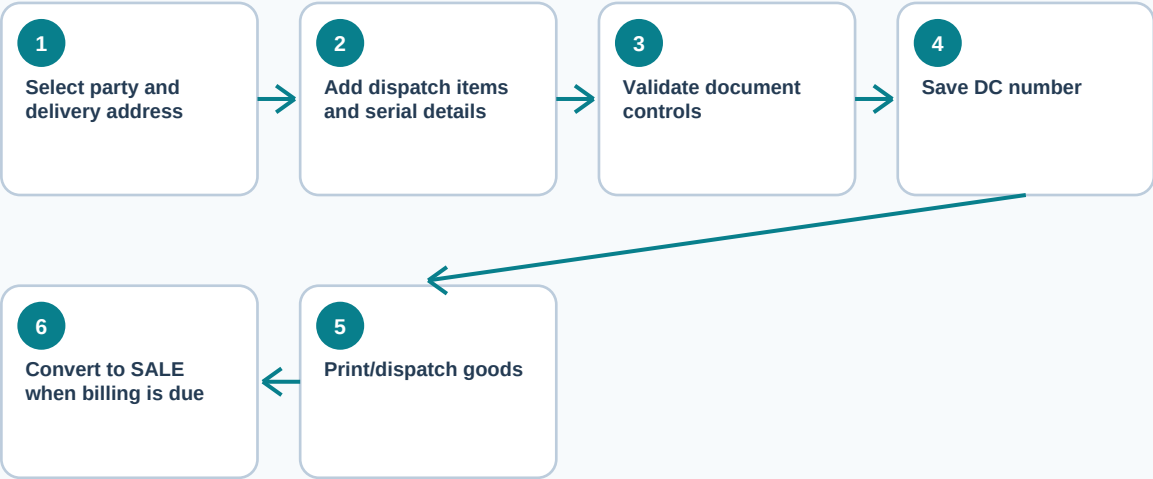


ACCOUNTING EFFECT None until converted to SALE	STOCK EFFECT None until converted to SALE	MAIN RECORDS bills(format=QUOT), bill_sequences, audit_log	KEY CONTROLS FY numbering, party snapshot, operator edit reason
--	---	--	---

Delivery Challan Transaction

ACTORS Owner, counter

PURPOSE Records goods delivered or dispatched with delivery-party details and can later become the linked sale invoice.

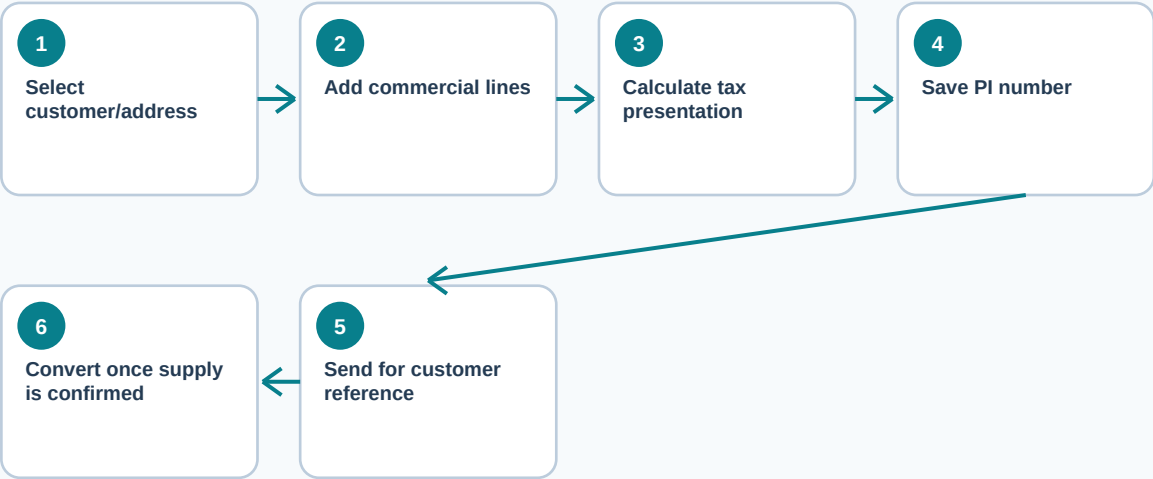


ACCOUNTING EFFECT	STOCK EFFECT	MAIN RECORDS	KEY CONTROLS
None until converted	Current implementation posts stock on resulting SALE	bills(format=DC), bill_sequences, audit_log	Document number never reused; stable delivery snapshot

Proforma Invoice Transaction

ACTORSOwner, finance, counter

PURPOSECreates a non-accounting advance invoice document that can later be converted into the final sale invoice.



ACCOUNTING EFFECT None until converted	STOCK EFFECT None until converted	MAIN RECORDS bills(format=PI), bill_sequences, audit_log	KEY CONTROLS Conversion creates a new SALE and preserves source reference
--	---	--	---

Sale, POS and Project-Printing Invoice

ACTORS Owner, counter, permitted operator

PURPOSE Creates the fiscal sale document, posts tax and receivable/cash/bank journals, and reduces stock for product lines.

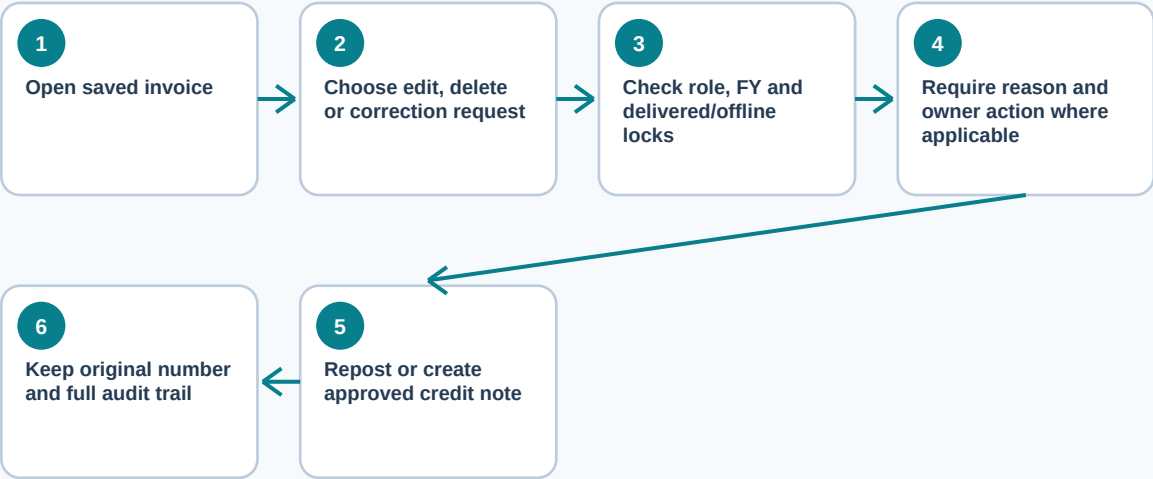


ACCOUNTING EFFECT	STOCK EFFECT	MAIN RECORDS	KEY CONTROLS
Debit cash/bank/receivable; credit sales and output GST	SALE quantity out for inventory items	bills, journal_entries, journal_lines, stock_movements, audit_log	Atomic save, FY lock, number non-reuse, org/party validation

Invoice Edit, Owner Delete and Invoice Correction

ACTORS Creator with reason; Owner for delete/approval

PURPOSE Controls changes to saved invoices, protects numbering and rebuilds the related accounting/stock effects safely.

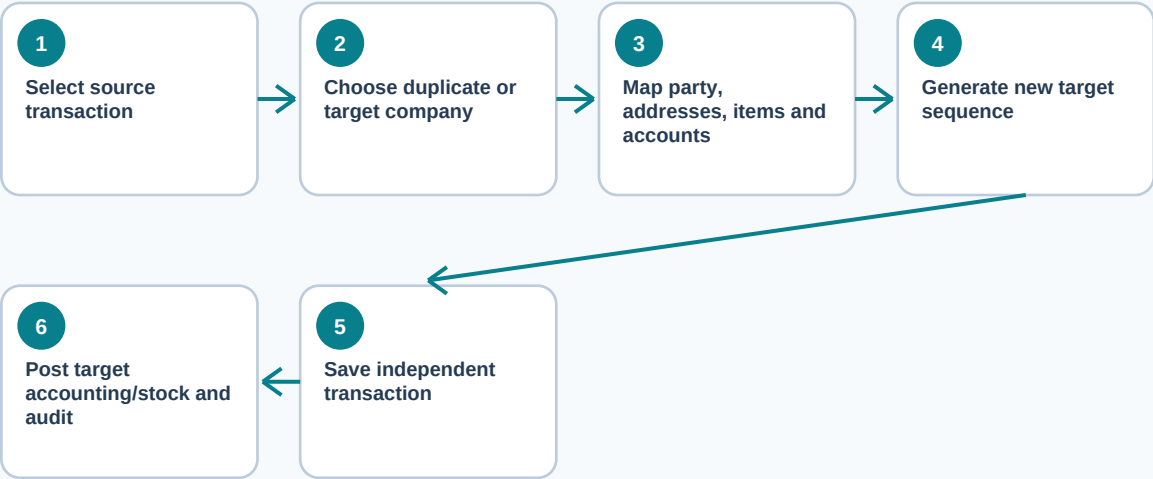


ACCOUNTING EFFECT Repost journal or post credit-note reversal	STOCK EFFECT Replace/remove movements or return stock where applicable	MAIN RECORDS bills, invoice_correction_requests, credit_debit_notes, audit_log	KEY CONTROLS Soft delete; mandatory reason; numbers never renumbered
---	--	--	--

Duplicate and Cross-Organization Voucher

ACTORS Owner, permitted staff

PURPOSE Copies an existing voucher as a new draft/transaction while requiring valid target-company party, address, item and account identities.

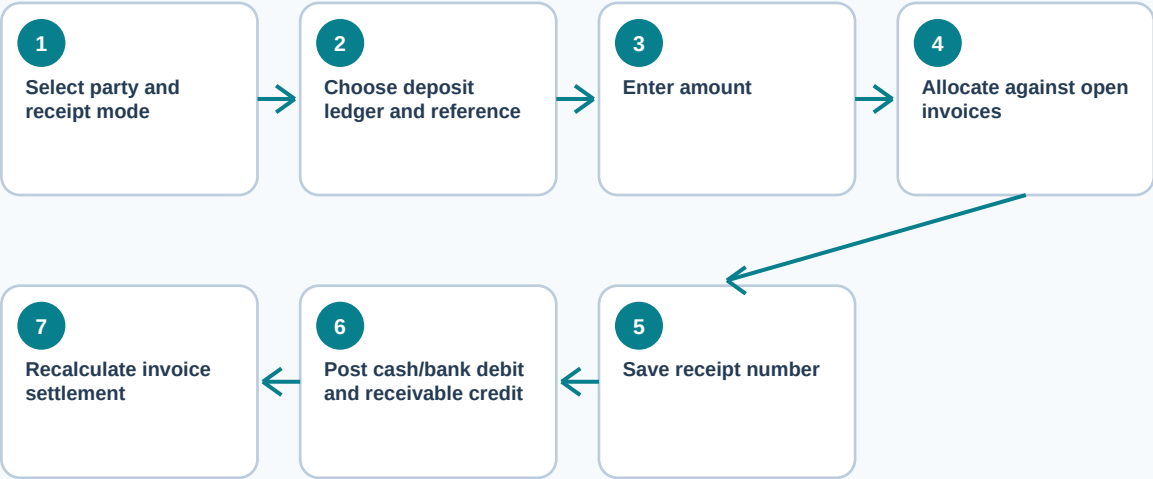


ACCOUNTING EFFECT	STOCK EFFECT	MAIN RECORDS	KEY CONTROLS
New posting only for accounting voucher types	New movement only for stock-changing types	bills/payments/purchases/expenses/notes/journals/PO, audit_log	No source ID reuse; cross-org mapping validation

Payment Received and Invoice Allocation

ACTORSOwner, finance, counter

PURPOSERecords money received, optionally into a personal/bank/cash ledger, and allocates it to one or more invoices.

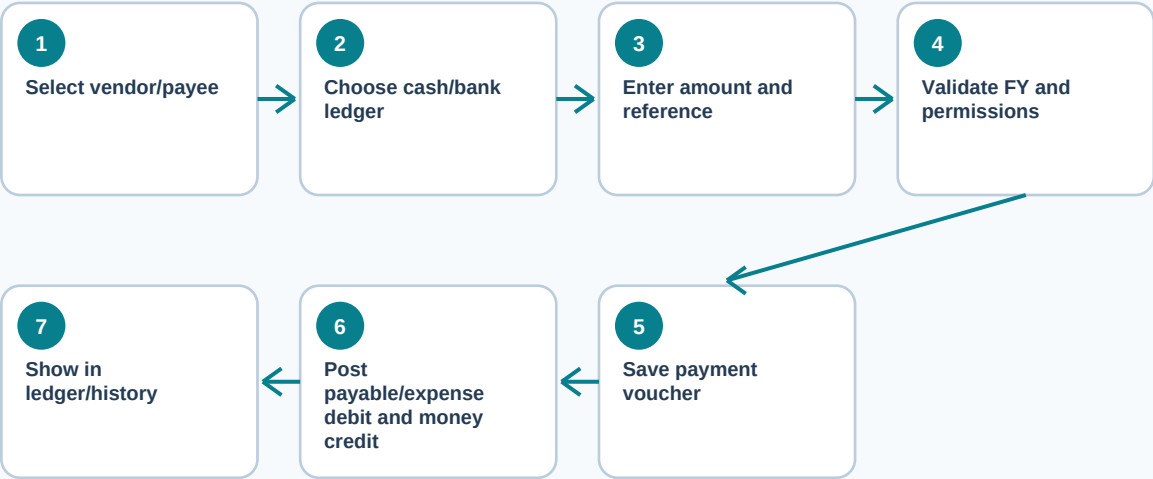


ACCOUNTING EFFECT Debit selected cash/bank; credit receivable or other income	STOCK EFFECT None	MAIN RECORDS payments(type=received), payment_allocations, journal_entries	KEY CONTROLS Allocation cannot exceed valid amount; FY/shift controls
---	-----------------------------	--	---

Payment Paid / Expense Settlement

ACTORS Owner, finance

PURPOSE Records outgoing cash/bank payment to a vendor or general expense destination.

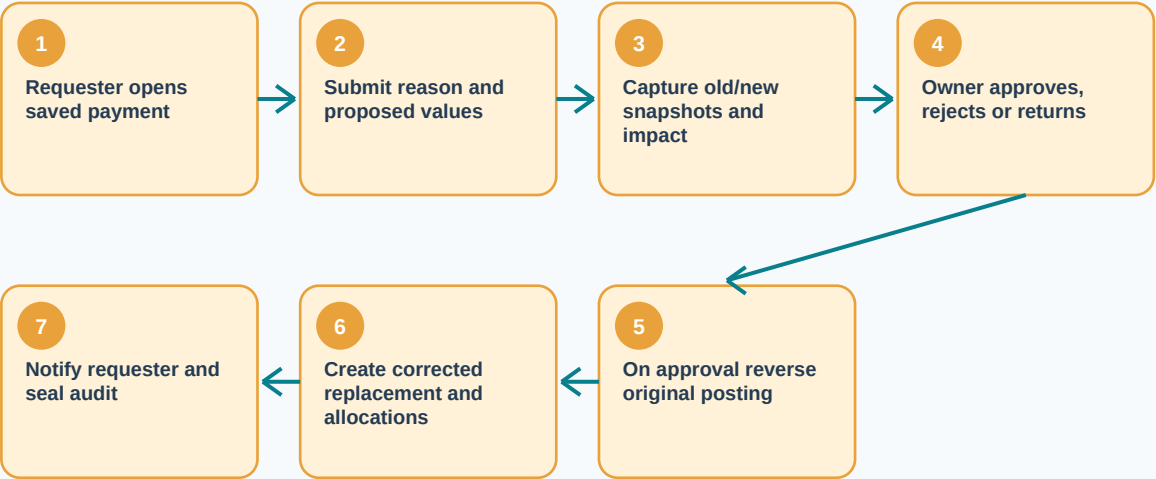


ACCOUNTING EFFECT	STOCK EFFECT	MAIN RECORDS	KEY CONTROLS
Debit payable/general expense; credit cash/bank	None	payments(type=paid), journal_entries, journal_lines	Owner/finance permission, immutable number, audit

Owner Approval for Payment Corrections

ACTORSRequester: operator/counter; Approver: owner

PURPOSEProposed controlled correction that preserves the original payment and applies a reversal plus corrected replacement only after owner approval.

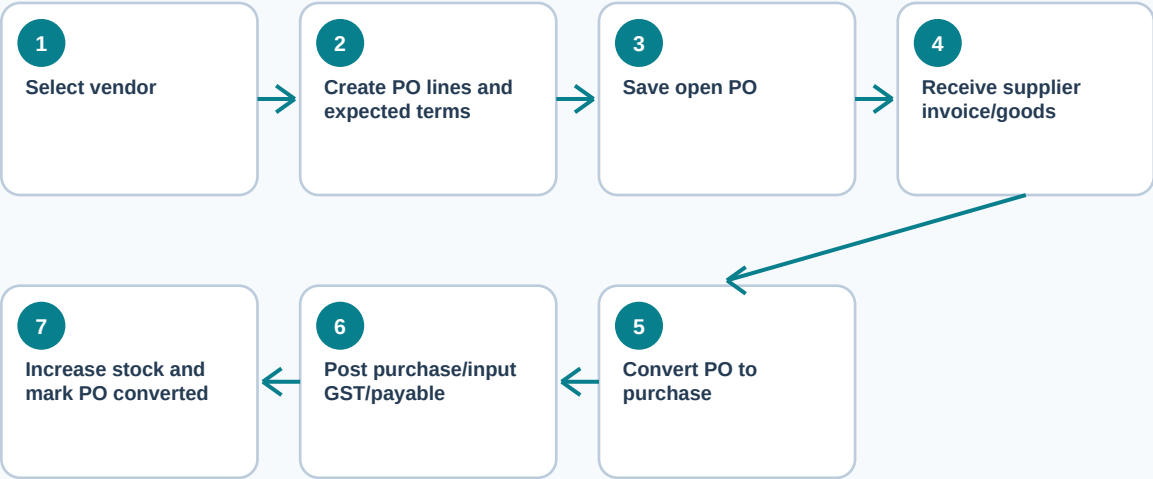


ACCOUNTING EFFECT Reversal plus replacement; never silent in-place overwrite	STOCK EFFECT None	MAIN RECORDS Proposed payment_correction_requests plus existing payment/journal tables	KEY CONTROLS Owner re-authentication, idempotency, closed-FY exception handling
--	-----------------------------	--	---

Purchase Order to Purchase Voucher

ACTORSOwner, purchase/finance staff

PURPOSECreates a vendor order without accounting impact, then converts accepted supply into a purchase voucher.

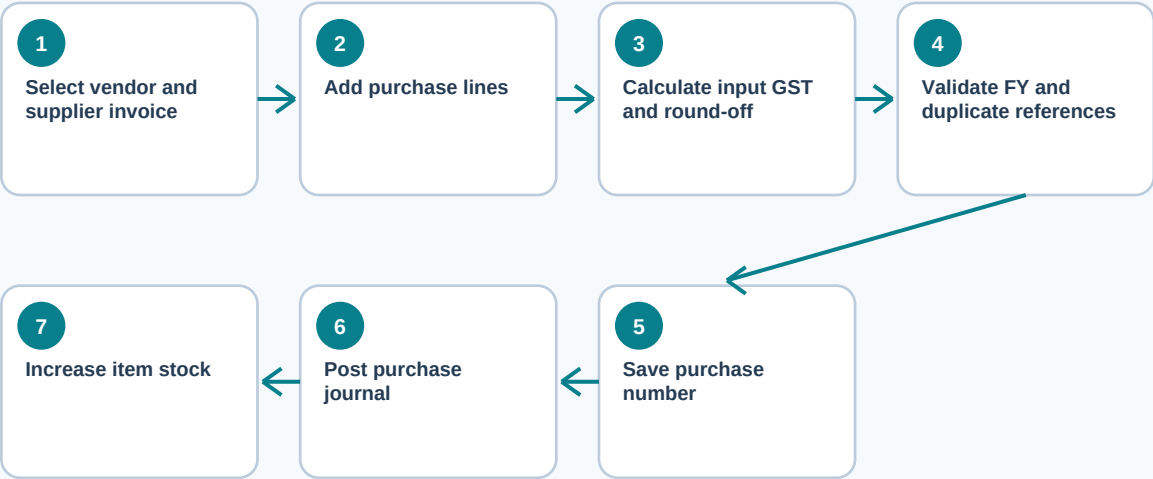


ACCOUNTING EFFECT	STOCK EFFECT	MAIN RECORDS	KEY CONTROLS
At conversion: debit purchase/input GST; credit payable/cash/bank	Purchase quantity in	purchase_orders, purchases, journal_entries, stock_movements	Only open PO converts; new purchase number generated

Direct Purchase Voucher

ACTORS Owner, finance/purchase staff

PURPOSE Records supplier goods/services, input tax, payable or immediate payment and inventory receipt.

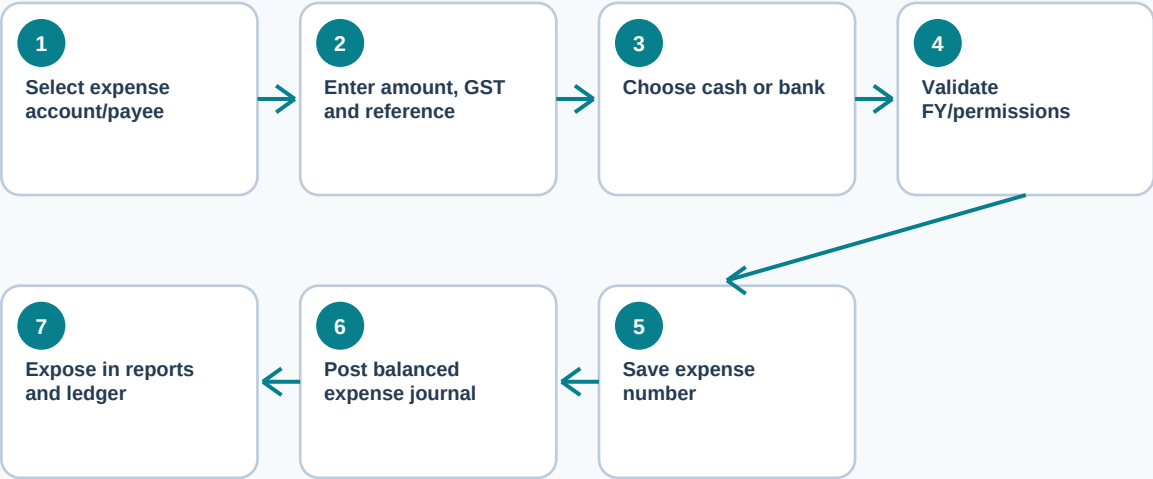


ACCOUNTING EFFECT Debit purchases/input GST; credit payable/cash/bank	STOCK EFFECT Quantity in for inventory lines	MAIN RECORDS purchases, journal_entries, journal_lines, stock_movements	KEY CONTROLS Organization, date/FY, vendor and item validation
---	--	---	--

Expense Voucher

ACTORS Owner, finance

PURPOSE Records operating expense, optional input GST and the selected cash/bank payment source.

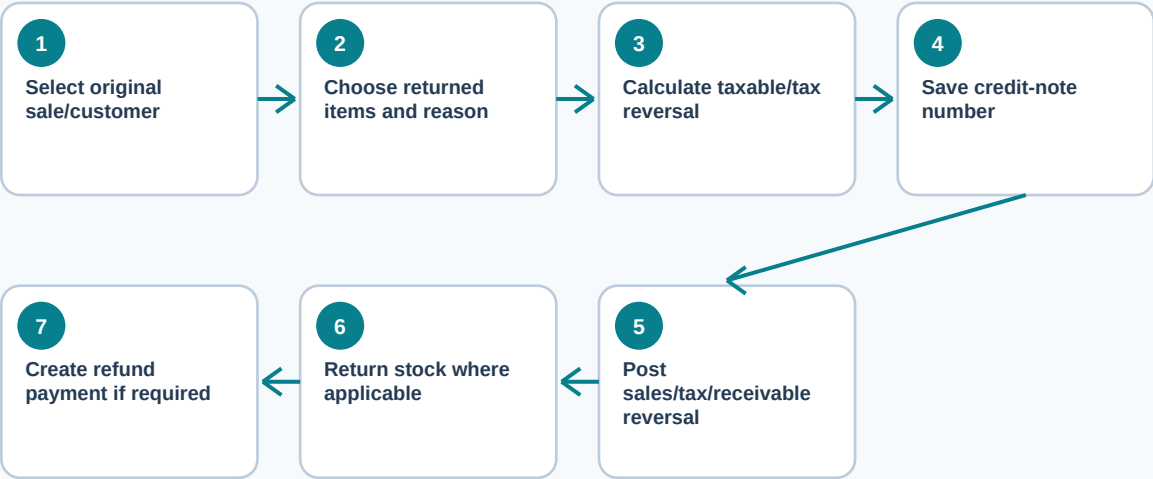


ACCOUNTING EFFECT Debit expense/input GST; credit cash/bank	STOCK EFFECT None	MAIN RECORDS expenses, journal_entries, journal_lines, audit_log	KEY CONTROLS Balanced posting, FY lock, role permission
---	-----------------------------	--	---

Credit Note, Sales Return and Refund

ACTORSOwner, finance

PURPOSEReduces a customer sale, tax and outstanding amount; optional refund payment and stock return are linked to the original invoice.

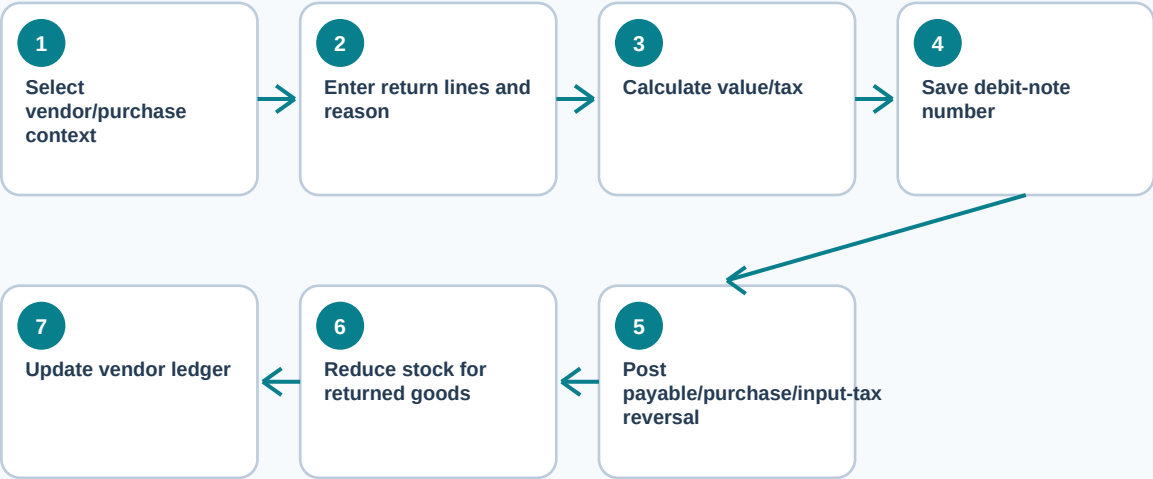


ACCOUNTING EFFECT	STOCK EFFECT	MAIN RECORDS	KEY CONTROLS
Debit sales/output tax; credit receivable; optional refund posting	Quantity in for returned inventory	credit_debit_notes(note_type=credit), payments, stock_movements	Original bill link, amount validation, owner controls

Debit Note and Purchase Return

ACTORS Owner, finance

PURPOSE Records goods/value returned to a supplier and reduces purchase/input-tax or payable effects.

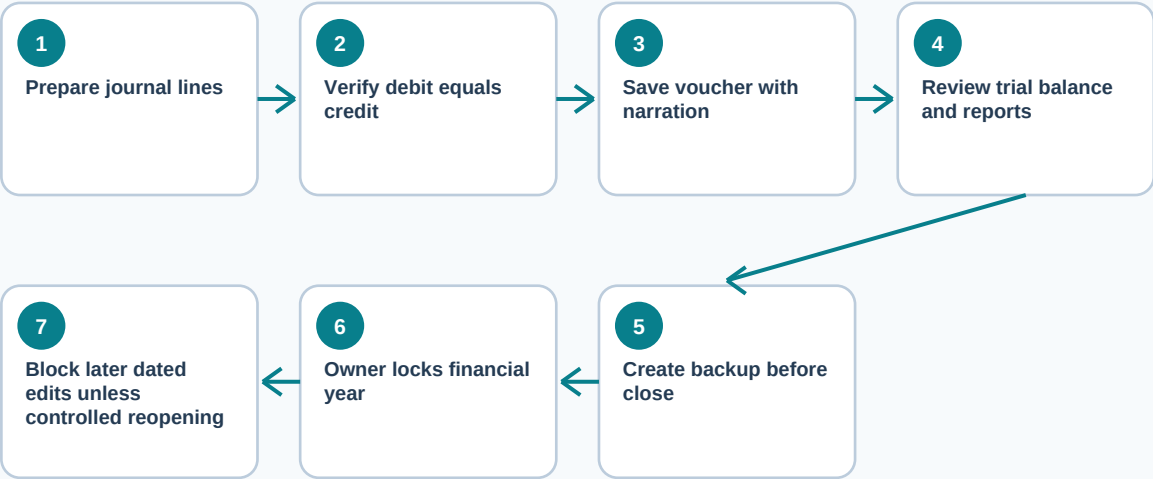


ACCOUNTING EFFECT Debit payable; credit purchases/input GST	STOCK EFFECT Quantity out for purchase return lines	MAIN RECORDS credit_debit_notes(note_type=debit), journal_entries, stock_movements	KEY CONTROLS FY lock, balanced journal, audit and number non-reuse
---	---	---	--

Manual Journal and Financial-Year Closing

ACTORS Owner, accountant

PURPOSE Posts controlled debit/credit adjustments and locks completed financial years against later transaction changes.

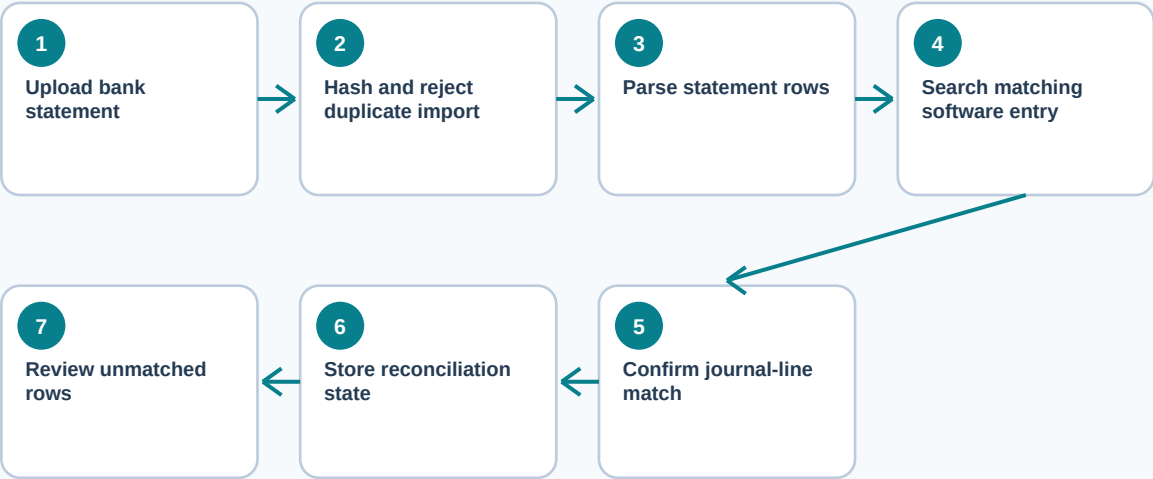


ACCOUNTING EFFECT	STOCK EFFECT	MAIN RECORDS	KEY CONTROLS
Direct balanced journal and year-end adjustments	None unless separately rebuilt from source documents	journal_entries, journal_lines, financial_year_locks, backup_log	Owner/accountant only, balanced totals, lock audit

Bank Statement Import and Reconciliation

ACTORSOwner, accountant

PURPOSEImports XLSX/CSV statement rows with duplicate protection and matches each row to a recorded journal line.

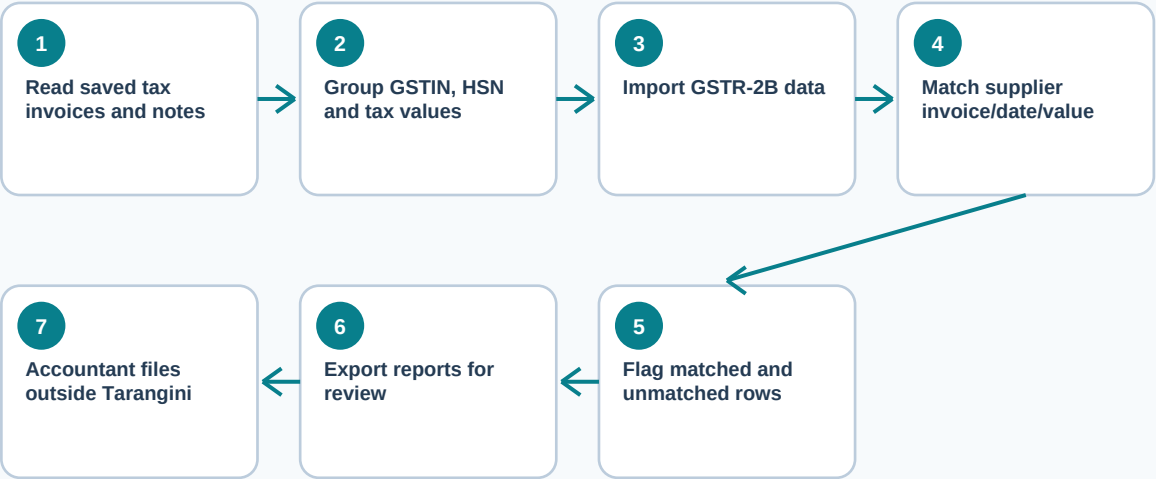


ACCOUNTING EFFECT	STOCK EFFECT	MAIN RECORDS	KEY CONTROLS
No new posting from match itself	None	bank_statement_imports, bank_statement_rows, bank_reconciliation	File hash, organization scope, reversible match audit

GST Reports and GSTR-2B Matching

ACTORSOwner, accountant

PURPOSEBuilds sales GST summaries and compares imported GSTR-2B purchase rows with Tarangini purchase records.

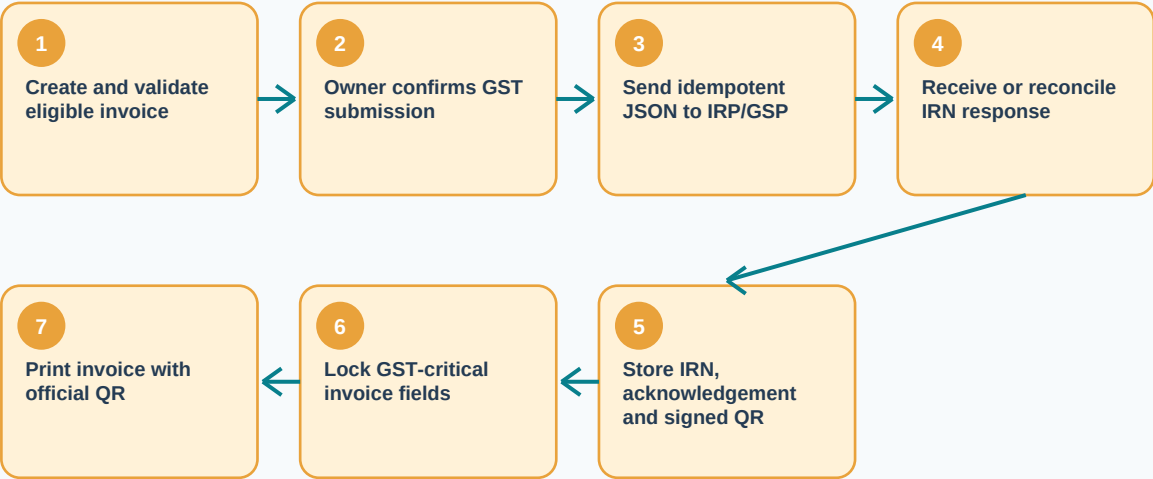


ACCOUNTING EFFECT	STOCK EFFECT	MAIN RECORDS	KEY CONTROLS
Reports existing postings; no automatic filing journal	None	bills, purchases, credit_debit_notes, gstr2b_imports/rows	GSTIN/date/value validation and accountant review

GST E-Invoice Registration

ACTORSOwner/finance, IRP or authorized GSP

PURPOSEProposed online submission of an eligible saved invoice to obtain IRN, acknowledgement and signed QR data.

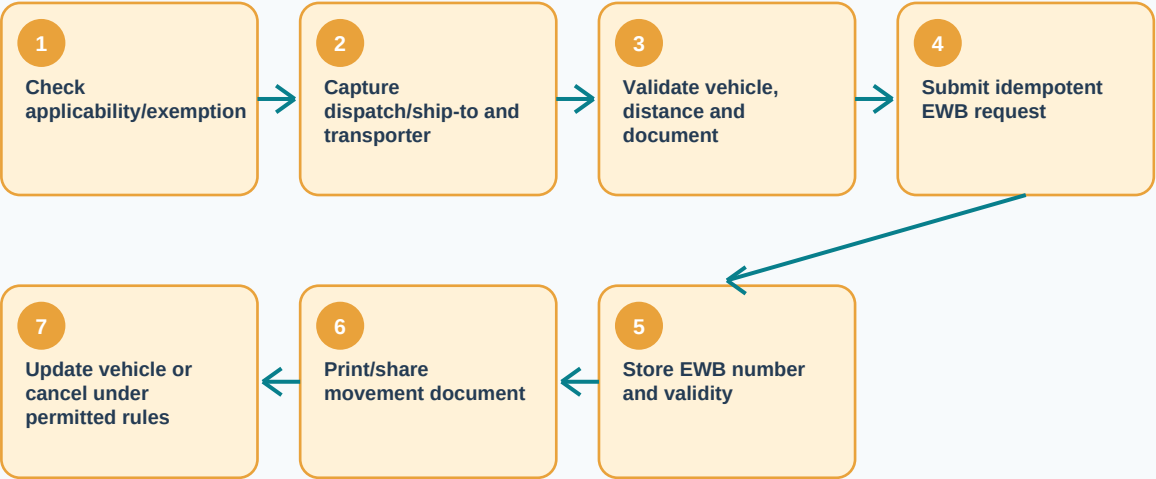


ACCOUNTING EFFECT Uses existing saved invoice posting	STOCK EFFECT Uses existing invoice stock movement	MAIN RECORDS Proposed GST submission/request/response records linked to bill_id	KEY CONTROLS Online only, encrypted credentials, duplicate reconciliation, cancellation rules
---	---	---	---

GST E-Way Bill Generation and Cancellation

ACTORSOwner/dispatch, transporter, E-Way Bill API

PURPOSEProposed goods-movement process using invoice/challan, dispatch, transporter, vehicle and distance details.

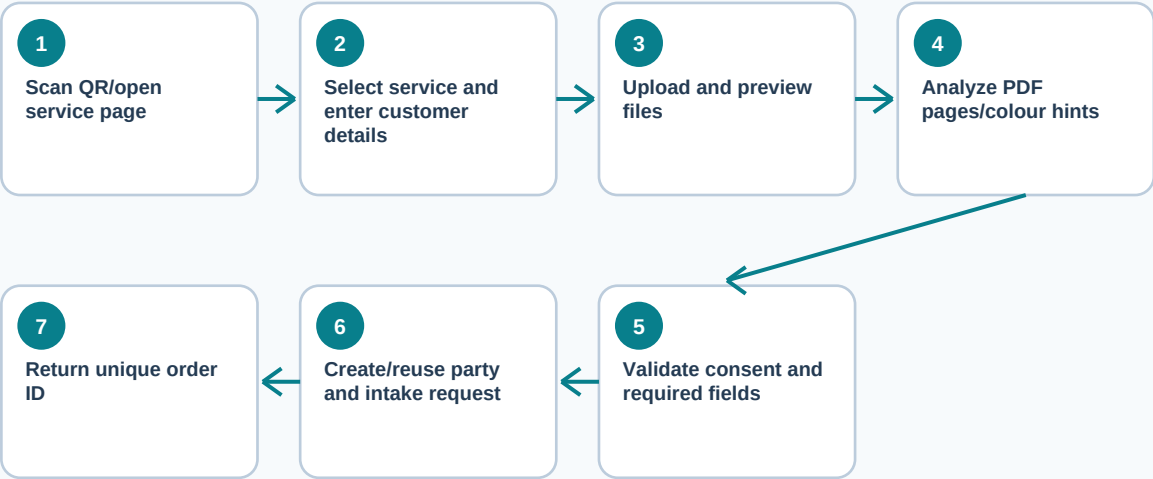


ACCOUNTING EFFECT No additional journal	STOCK EFFECT References existing dispatch/sale movement	MAIN RECORDS Proposed EWB request/response/history linked to bill/DC	KEY CONTROLS Online API, state/job-work rules, expiry and cancellation audit
---	---	--	--

Customer QR Intake, Upload and PDF Analysis

ACTORSCustomer, owner/counter

PURPOSEAccepts a LAN customer request, contact details, service selection, consent and files, then returns a unique order ID.

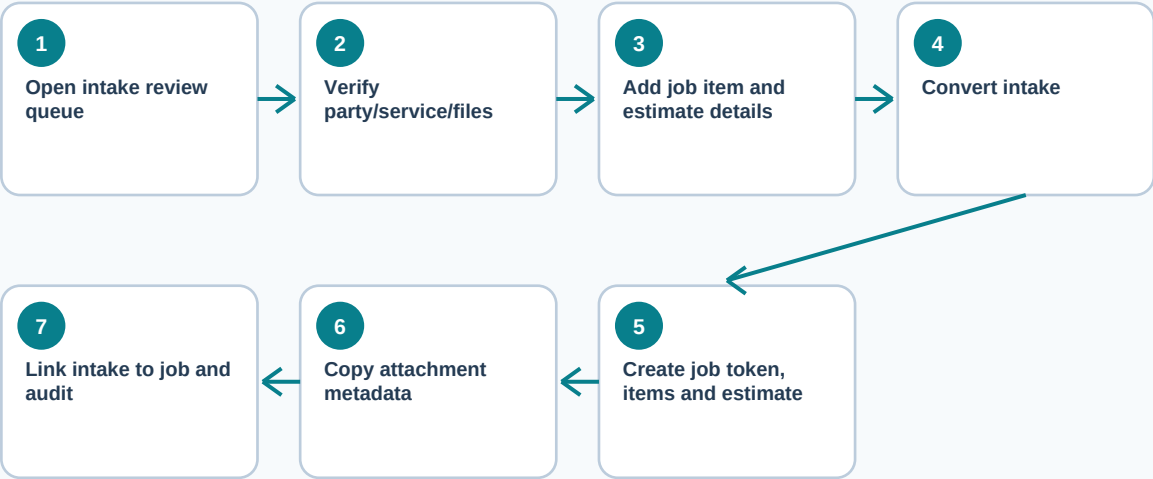


ACCOUNTING EFFECT	STOCK EFFECT	MAIN RECORDS	KEY CONTROLS
None	None	job_intake_requests, job_intake_attachments, parties, analysis jobs	Idempotency key, file limits, metadata/hash, required-field focus

Intake Review and Job Creation

ACTORSOwner, counter

PURPOSEReviews the customer submission, corrects commercial details and converts it once into a uniquely numbered job order.

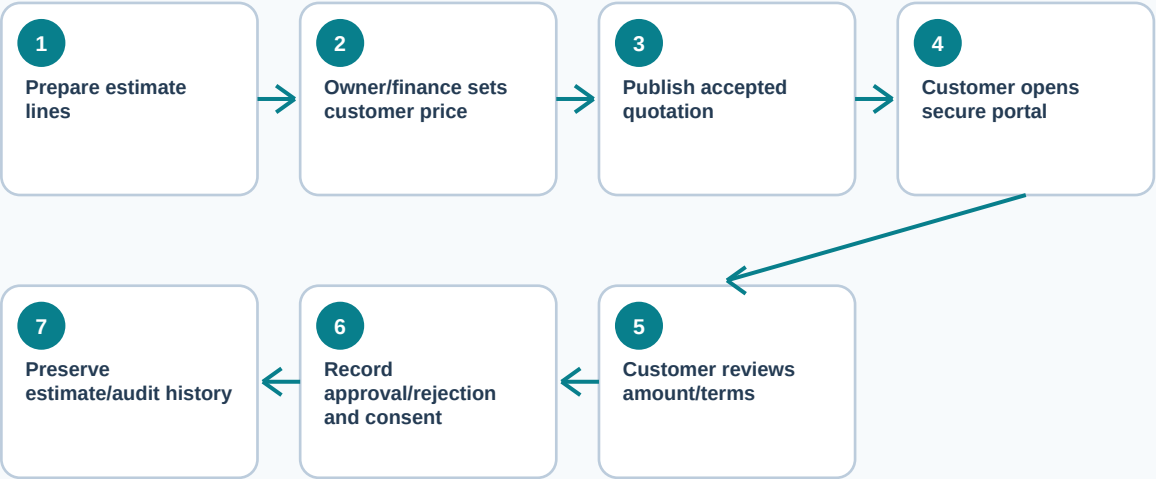


ACCOUNTING EFFECT None until job billing	STOCK EFFECT None until material consumption	MAIN RECORDS job_orders, job_items, job_estimates, job_attachments	KEY CONTROLS One conversion per intake; organization and party linkage
--	--	--	--

Estimate, Quotation and Customer Approval

ACTORSOwner/finance, customer

PURPOSEStores job estimate versions and exposes the accepted quotation and later additional-work decisions to the customer portal.

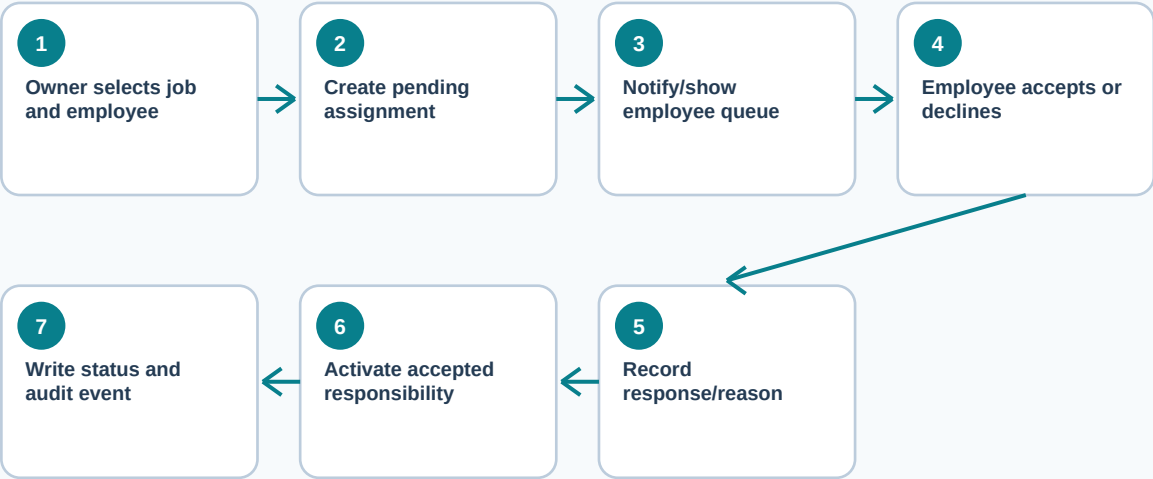


ACCOUNTING EFFECT None until first/final bill	STOCK EFFECT None	MAIN RECORDS job_estimates, job_customer_approvals, job_audit_events	KEY CONTROLS Version/history, secure token, immutable decision evidence
---	-----------------------------	--	---

Job Assignment and Operator Response

ACTORS Owner/senior operator, assigned employee

PURPOSE Routes a job to the responsible operator/engineer and records acceptance, decline, handoff and instructions.



ACCOUNTING EFFECT	STOCK EFFECT	MAIN RECORDS	KEY CONTROLS
None	None	job_assignments, job_status_events, job_audit_events	Assigned-user visibility, delivered-job lock, response history

Production Status, Work Report and Daily Log

ACTORSOperator, engineer, owner

PURPOSETracks job progress, work performed, pending work, file names and miscellaneous employee activity through daily submission and owner review.

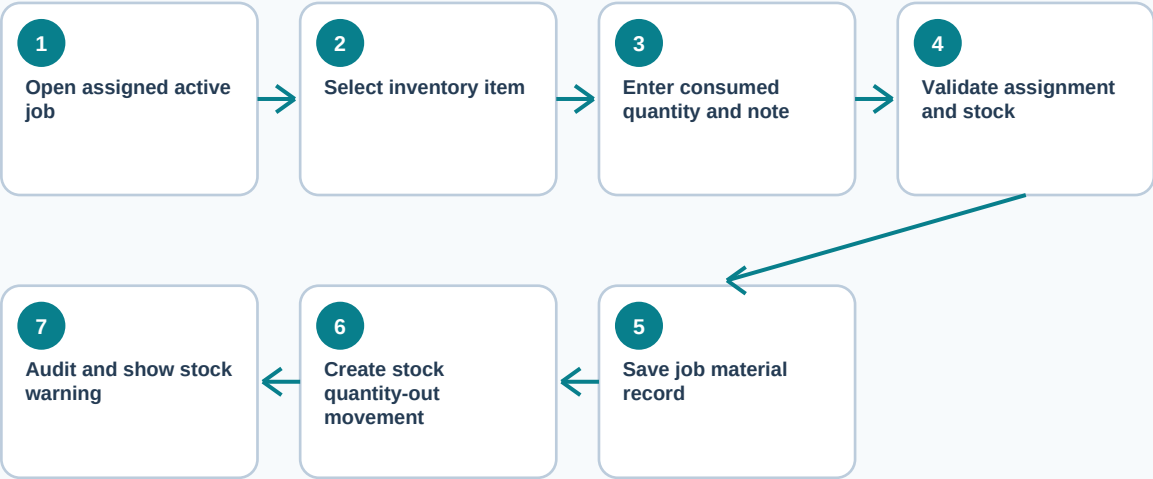


ACCOUNTING EFFECT None directly	STOCK EFFECT Only when separate material transaction is recorded	MAIN RECORDS job_status_events, job_work_reports, operator_daily_logs/entries	KEY CONTROLS Status transition rules, assignment check, submitted-log lock
---	--	---	--

Job Material Consumption

ACTORSAssigned operator/engineer

PURPOSELinks inventory consumed during production to the specific job and reduces available stock.

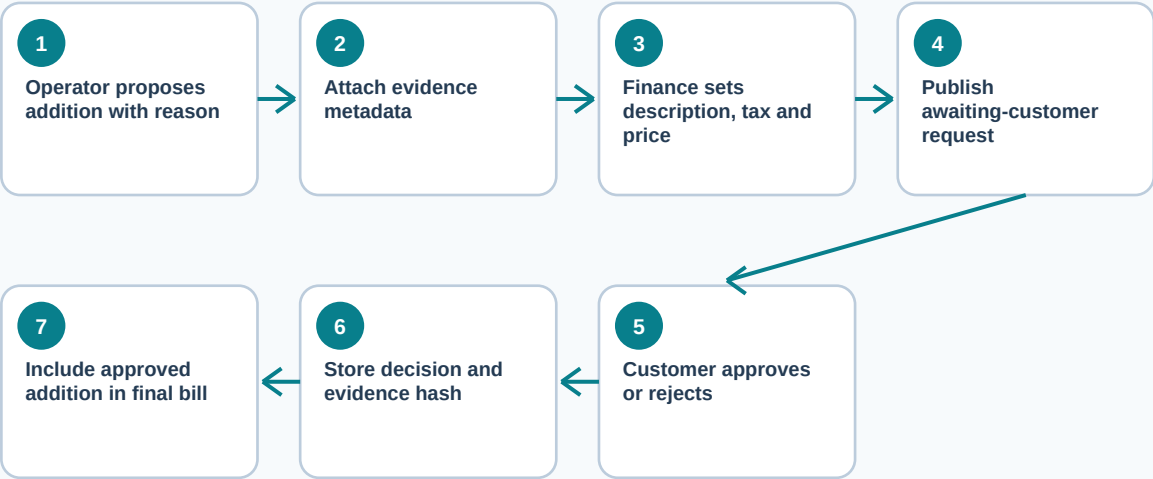


ACCOUNTING EFFECT No direct cost journal in current flow	STOCK EFFECT Quantity out linked to job material source	MAIN RECORDS job_material_consumptions, stock_movements, job_audit_events	KEY CONTROLS Accepted assignment, delivered-job lock, item/org validation
--	---	---	---

Additional Work, Evidence and Customer Decision

ACTORSOperator, finance/owner, customer

PURPOSECaptures unexpected additional work, prices it, requests customer approval and includes only approved value in final billing.

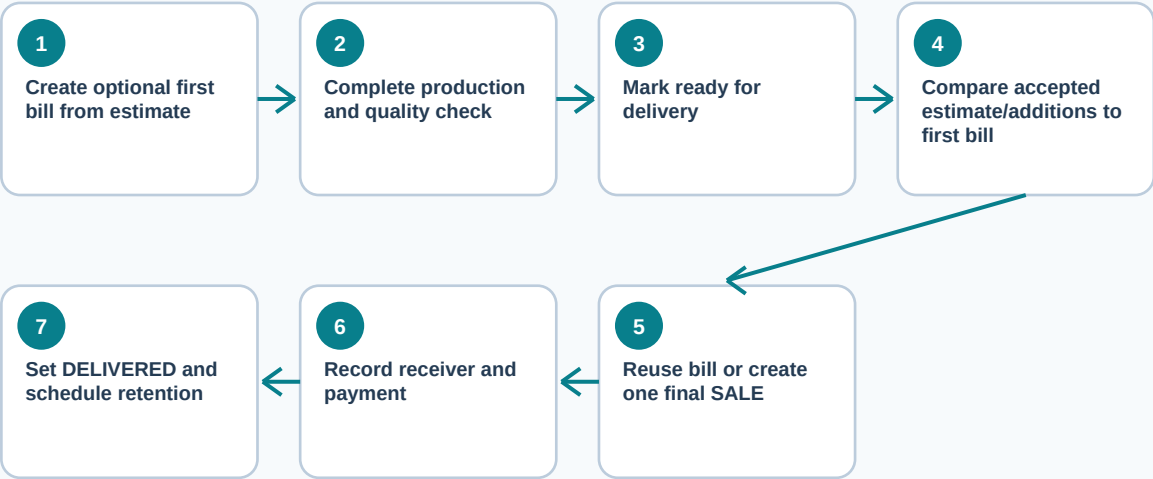


ACCOUNTING EFFECT None until included in bill	STOCK EFFECT Separate material records handle consumption	MAIN RECORDS job_additions, job_attachments, job_customer_approvals	KEY CONTROLS Price permission, secure customer token, decision audit
---	---	---	--

First Bill, Final Invoice, Delivery and Collection

ACTORSOwner, finance, counter, customer

PURPOSELinks job and invoice IDs, reuses a matching first bill, prevents duplicate final billing, records delivery and applies receipt payments.

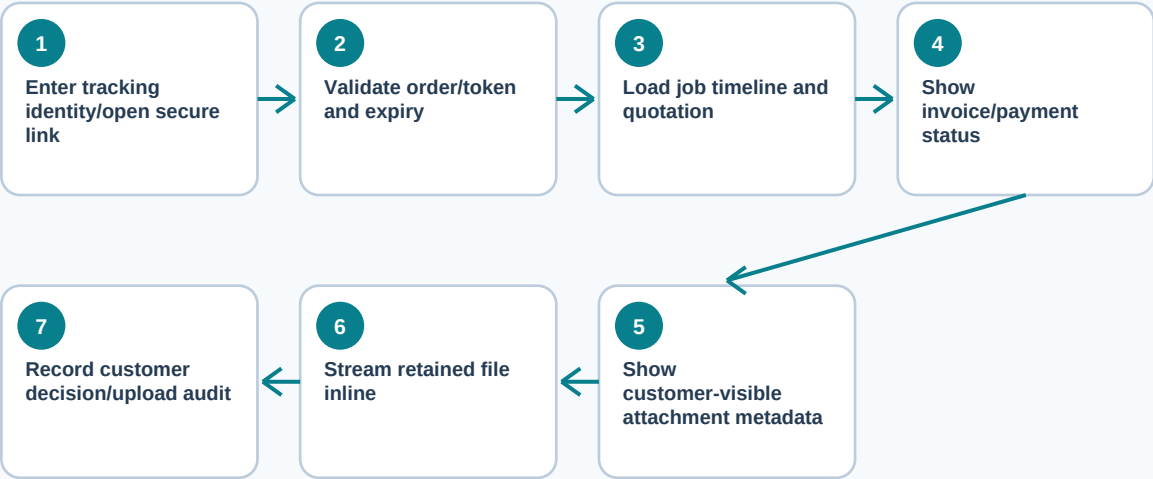


ACCOUNTING EFFECT Sale posting plus any receipt postings	STOCK EFFECT Invoice product lines/material transactions retain their movements	MAIN RECORDS job_orders, bills, payments, delivery acknowledgements, audit	KEY CONTROLS One final_bill_id, amount comparison, delivered-job lock
--	---	--	---

Order Tracking, Quotation and Output Preview

ACTORSCustomer

PURPOSELets the customer use order ID/secure token to see status, quotation, approvals, invoice/payment state and permitted inline previews.

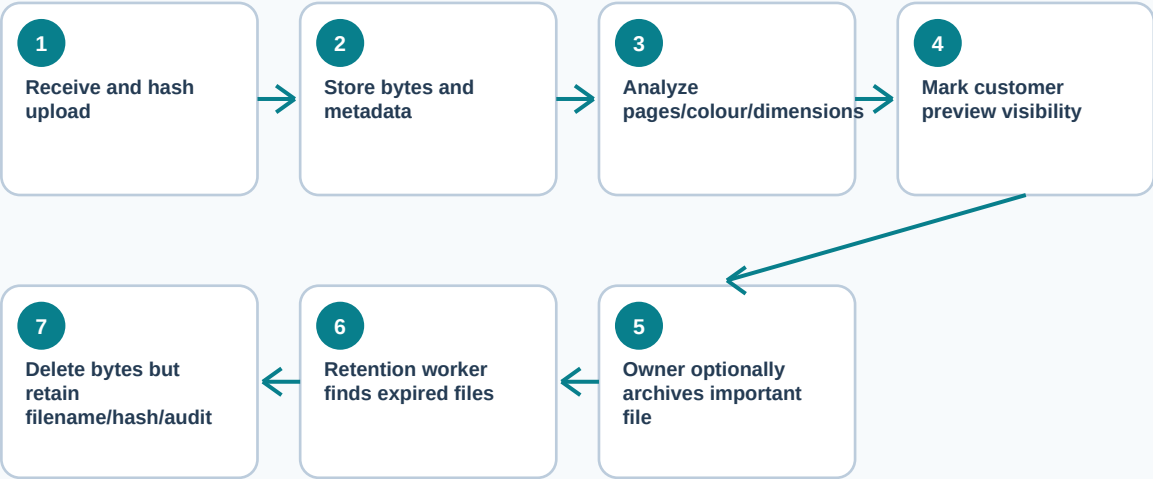


ACCOUNTING EFFECT Read-only visibility of linked invoice/payment state	STOCK EFFECT None	MAIN RECORDS job access tokens, job status, approvals, attachments, messages	KEY CONTROLS No cross-job access, no download button, visibility/retention checks
--	-----------------------------	--	---

Attachment Analysis, Archive and Retention

ACTORSCustomer, operator, owner, retention worker

PURPOSEStores controlled job files, retains metadata, analyzes PDF/images and deletes non-archived bytes seven days after completed delivery.

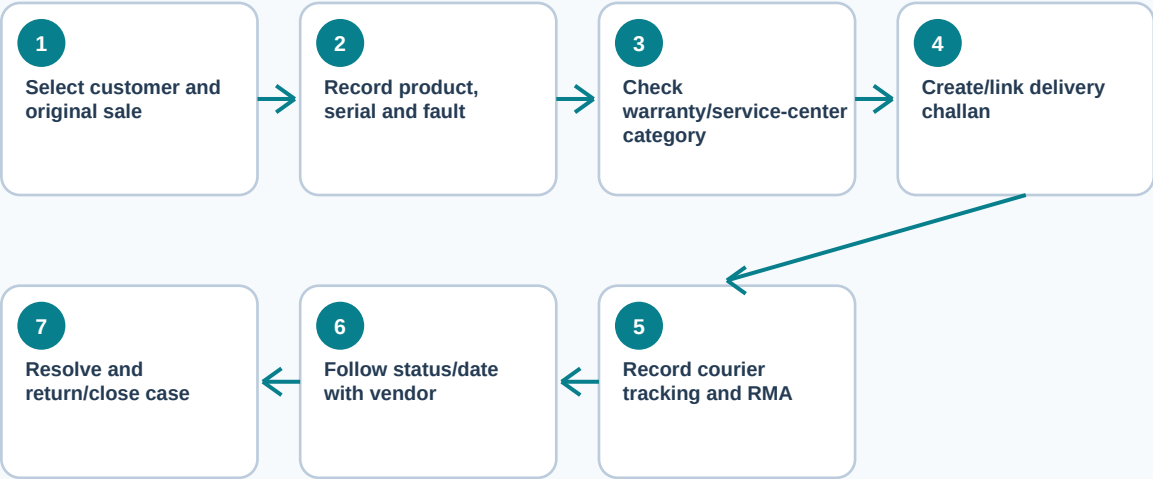


ACCOUNTING EFFECT	STOCK EFFECT	MAIN RECORDS	KEY CONTROLS
None	None	job_attachments, intake_attachments, attachment_analysis_jobs	File limits, path safety, archive exception, metadata retention

Warranty Replacement and Service-Center Tracking

ACTORS Owner, counter, operator, vendor/service center

PURPOSE Tracks a faulty sold product from customer receipt through delivery challan, courier/RMA follow-up and final resolution.

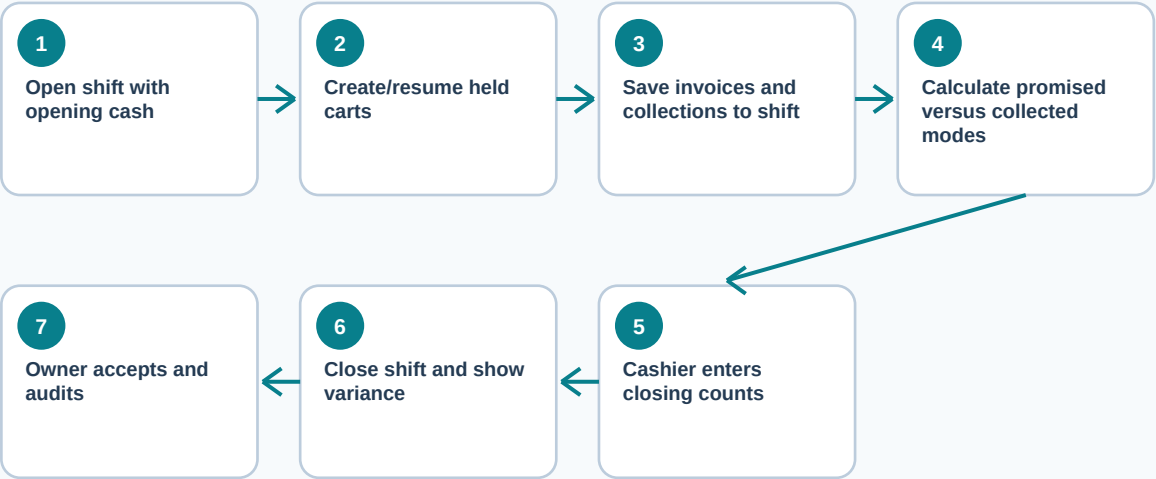


ACCOUNTING EFFECT No automatic journal; linked billing remains separate	STOCK EFFECT No automatic movement beyond linked transaction	MAIN RECORDS warranty_replacements, warranty_service_centers, bills, audit_log	KEY CONTROLS Unique WR number, party/bill validation, serial required
---	--	---	---

Shift, Held Bill and Day-End Acceptance

ACTORS Cashier, owner

PURPOSE Tracks counter opening, parked carts, shift-linked invoices/payments, counted cash, variance and owner acceptance.

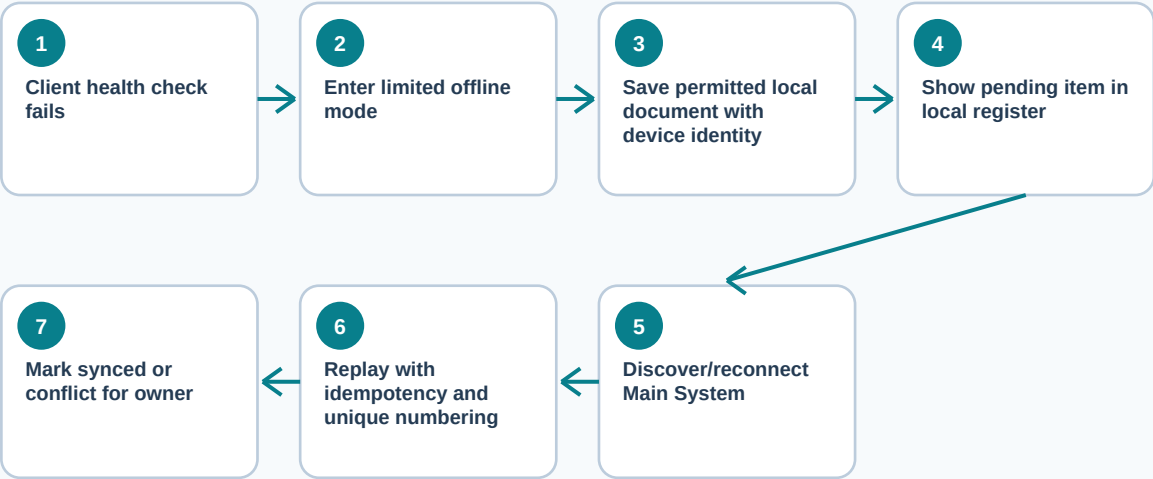


ACCOUNTING EFFECT Invoices/payments post normally; shift summarizes them	STOCK EFFECT Invoices post normal stock movements	MAIN RECORDS pos_shifts, held_bills, bills, payments	KEY CONTROLS One active shift rules, close/accept states, owner review
--	---	--	--

Client Offline Queue and Main-System Synchronization

ACTORSClient PC, Main System, owner

PURPOSEAllows approved document drafts such as quotations to queue locally, then synchronizes idempotently when the single Main System returns.



ACCOUNTING EFFECT Final accounting writes remain Main-System controlled	STOCK EFFECT Stock-changing offline writes remain restricted	MAIN RECORDS Client queue/cache plus server bills/audit after sync	KEY CONTROLS Stable device ID, retry idempotency, one Main System, conflict lock
---	--	--	--

Manual Migration Backup, Encrypted Backup and Restore

ACTORSOwner, automatic backup service

PURPOSECreates migration JSON or encrypted full-database backups, verifies history and rebuilds mappings/accounting during controlled restore.

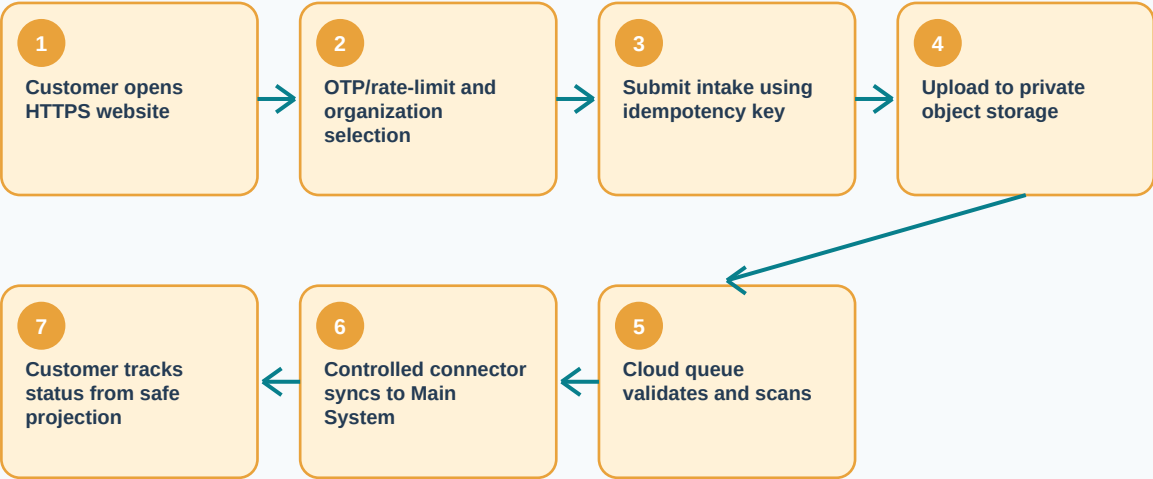


ACCOUNTING EFFECT All accounting data preserved/rebuilt	STOCK EFFECT All source movements preserved/rebuilt	MAIN RECORDS All business tables, backup_log, data_shift batches/maps	KEY CONTROLS Recovery key, separate disk, pre-upgrade backup, integrity check
---	---	---	---

Public Customer Website Transaction

ACTORSInternet customer, cloud API, Main System

PURPOSEProposed public HTTPS intake and tracking service that syncs controlled records without exposing SQLite or the LAN server to the internet.

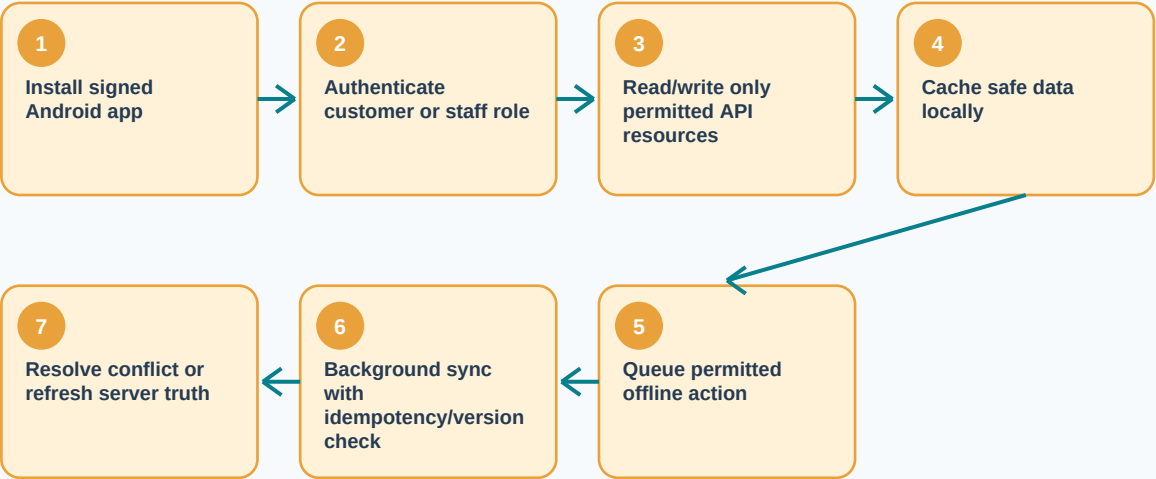


ACCOUNTING EFFECT No public direct accounting write in first rollout	STOCK EFFECT No public direct stock write	MAIN RECORDS Proposed cloud intake/upload/status projection plus mapped local IDs	KEY CONTROLS HTTPS, WAF/rate limit, org isolation, signed preview, conflict queue
--	---	---	---

Android Customer and Staff Application

ACTORSCustomer/staff app, public API, Main System

PURPOSEProposed Android client using the same versioned APIs as the website with an encrypted local cache and controlled background synchronization.



ACCOUNTING EFFECT Remote billing only after separate approval/security phase	STOCK EFFECT No unsafe offline stock writes	MAIN RECORDS Same API IDs: org, party, item, job, bill and payment references	KEY CONTROLS HTTPS, token rotation, device security, RBAC, no shared database
--	---	---	---